



Lattice-based Threshold Blind Signatures

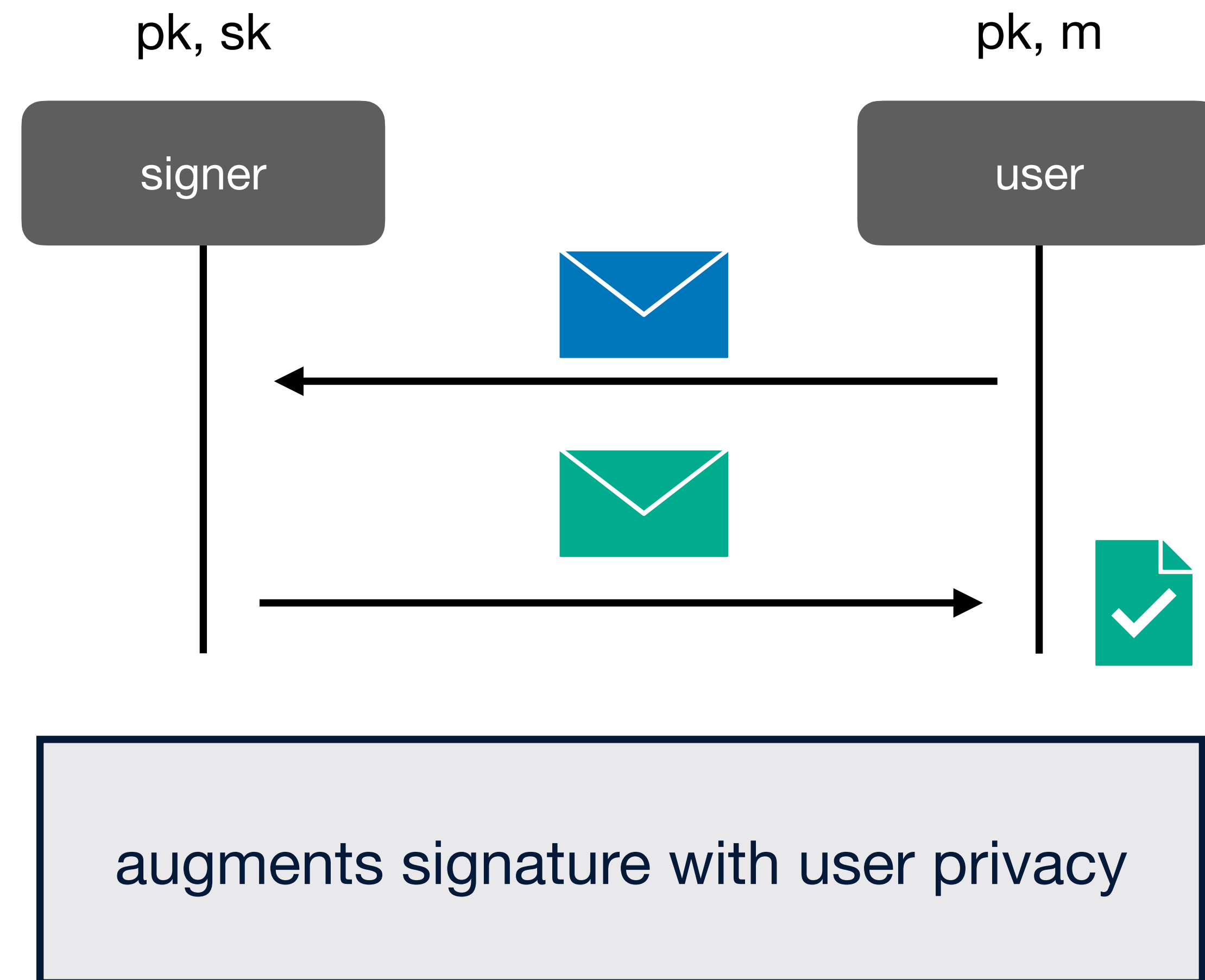
S&P 2026

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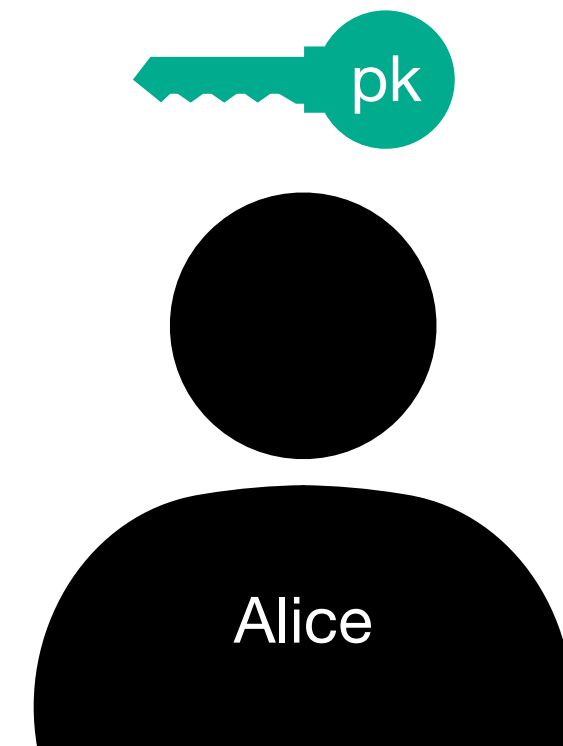
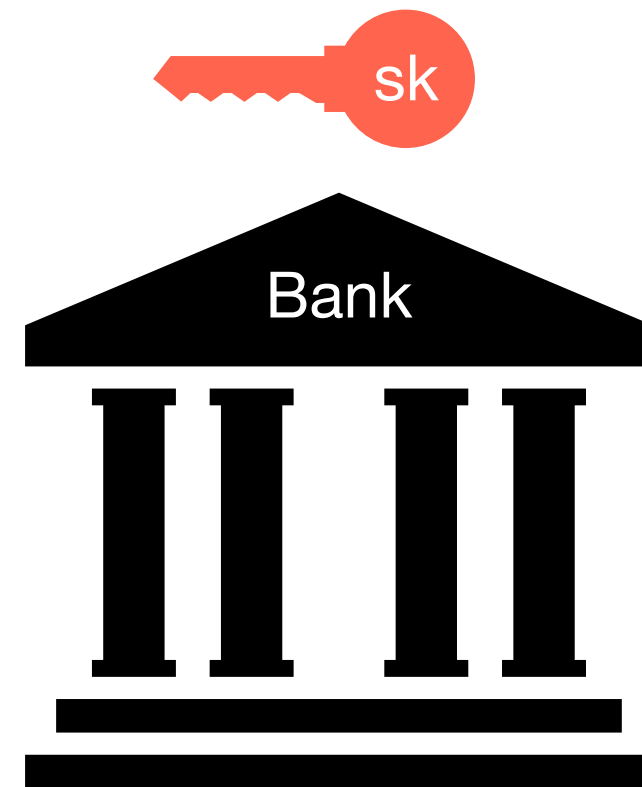
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Blind Signatures



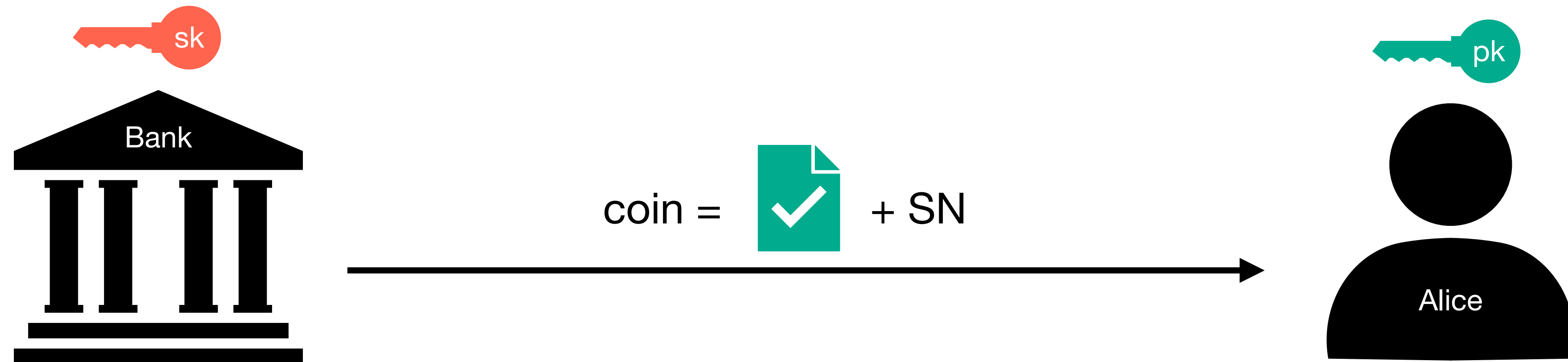
Applications

CBDCs



Applications

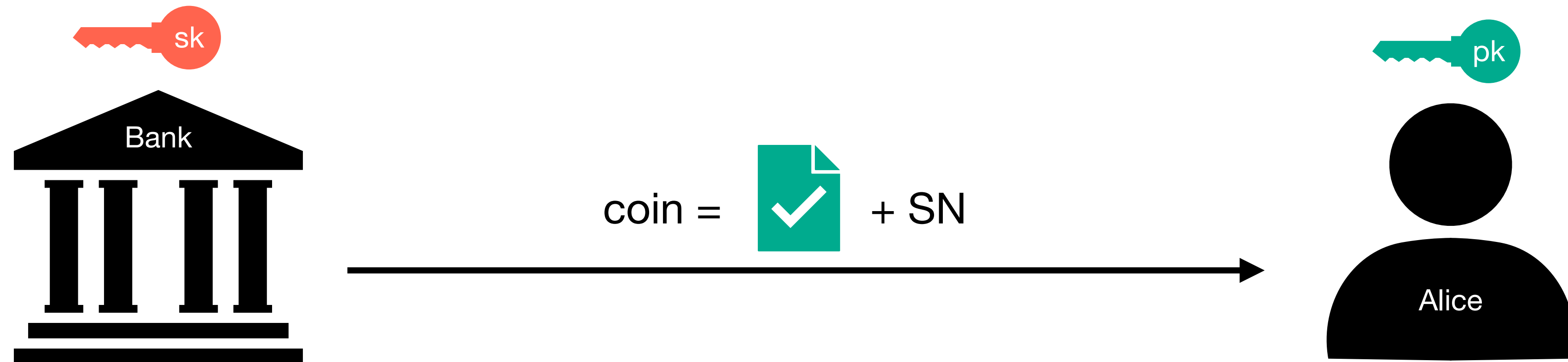
CBDCs



- SN is a *random* serial number chosen by Alice

Applications

CBDCs



- SN is a *random* serial number chosen by Alice
- The bank keeps track of used serial numbers

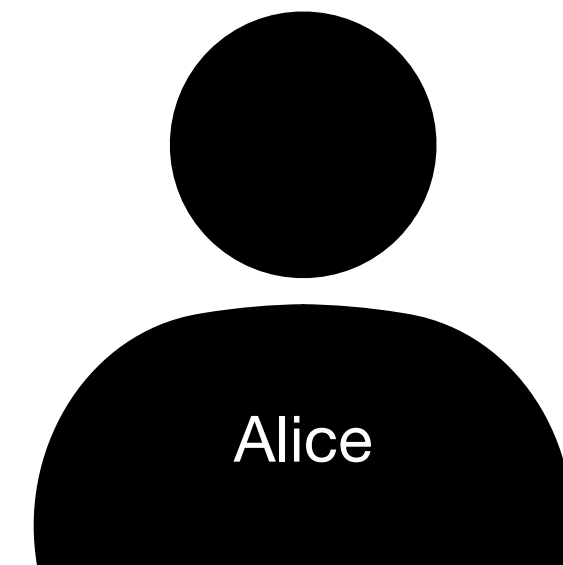
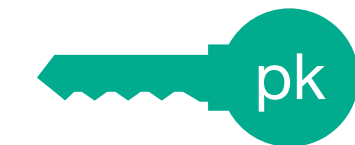
Applications

CBDCs

single point of failure



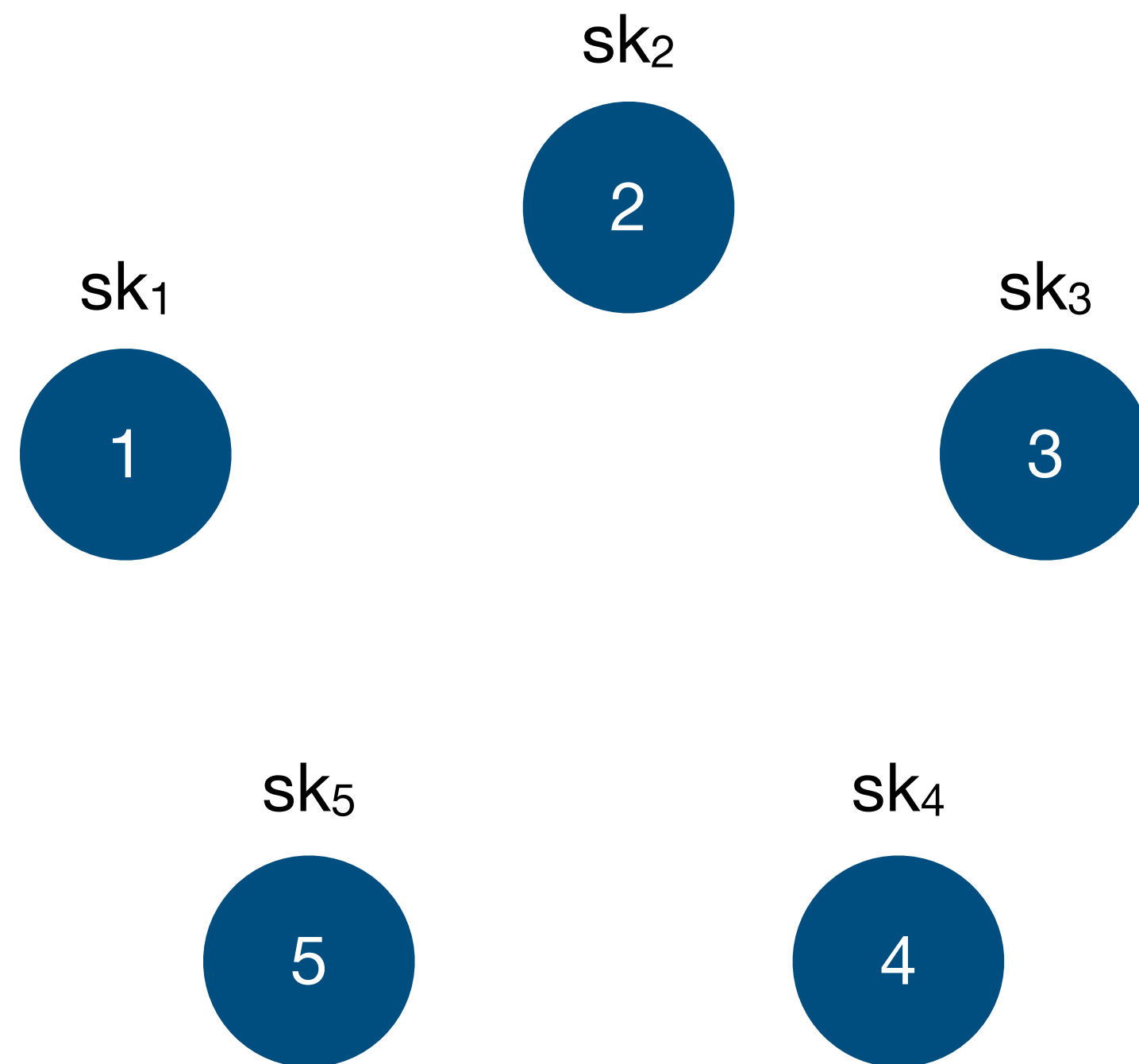
coin =  + SN



- SN is a *random* serial number chosen by Alice
- The bank keeps track of used serial numbers

Threshold Signatures

T-out-of-N

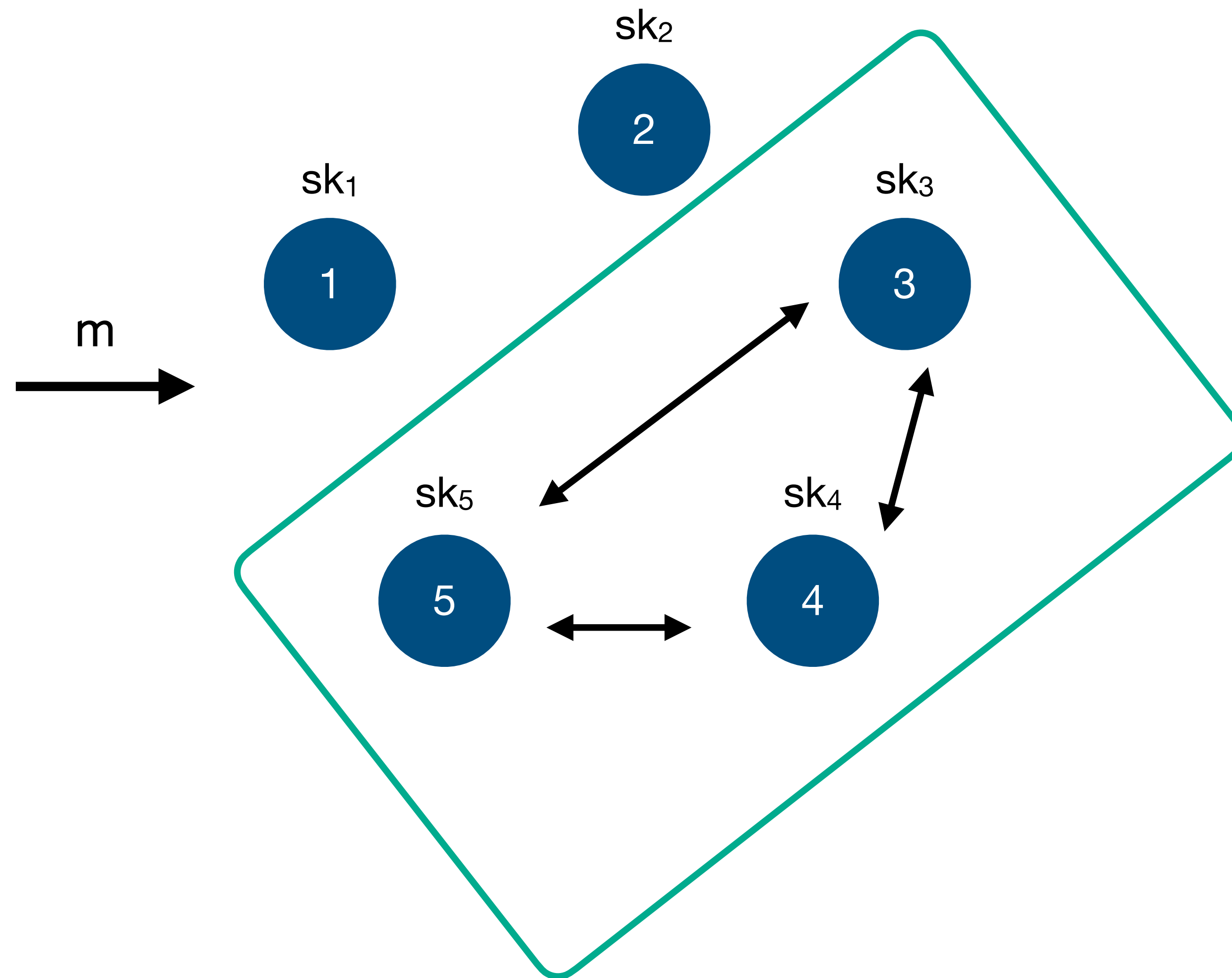


- **N**: number of shares
- **T**: threshold for issuance

Threshold Signatures

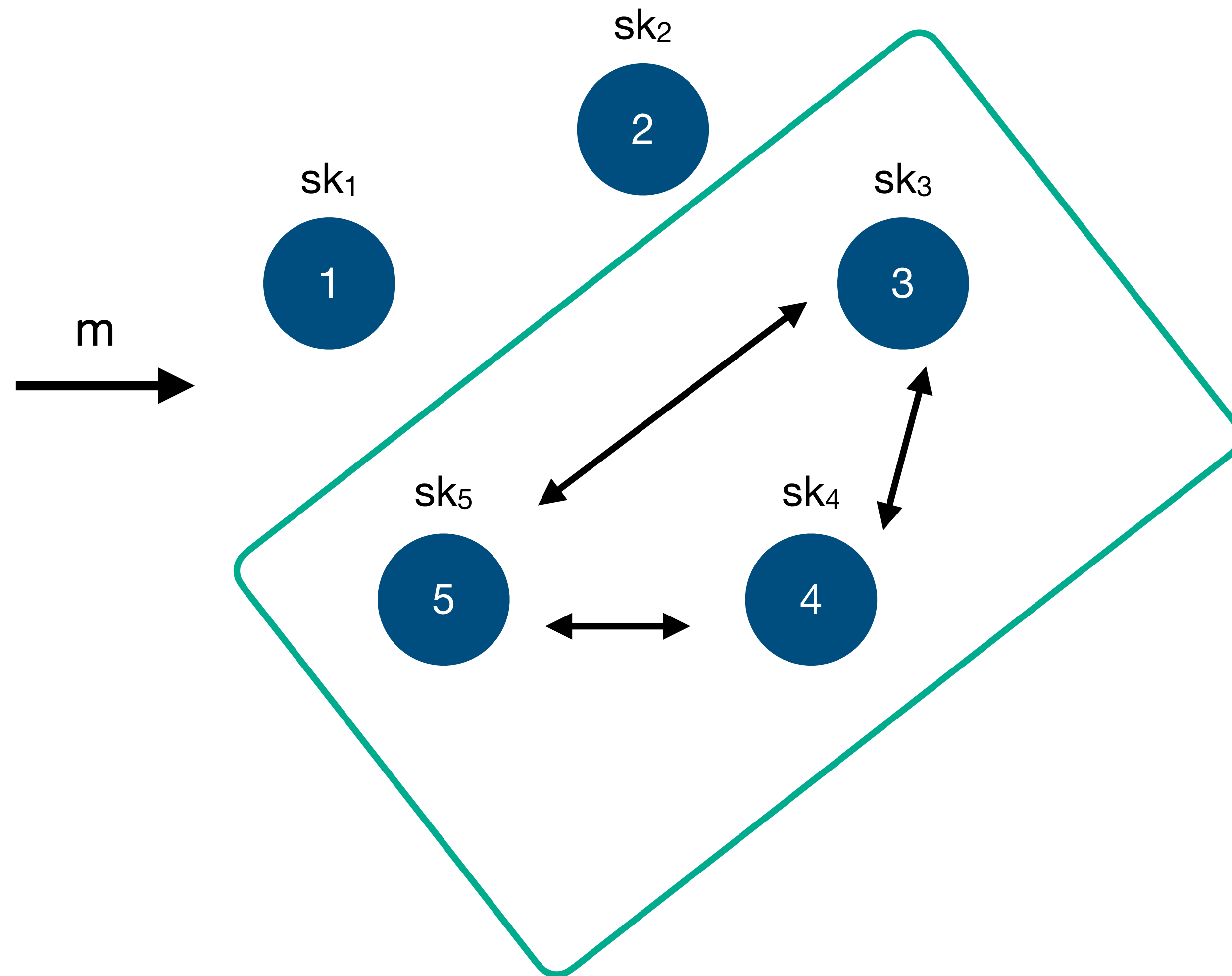
T-out-of-N

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Threshold Signatures

T-out-of-N



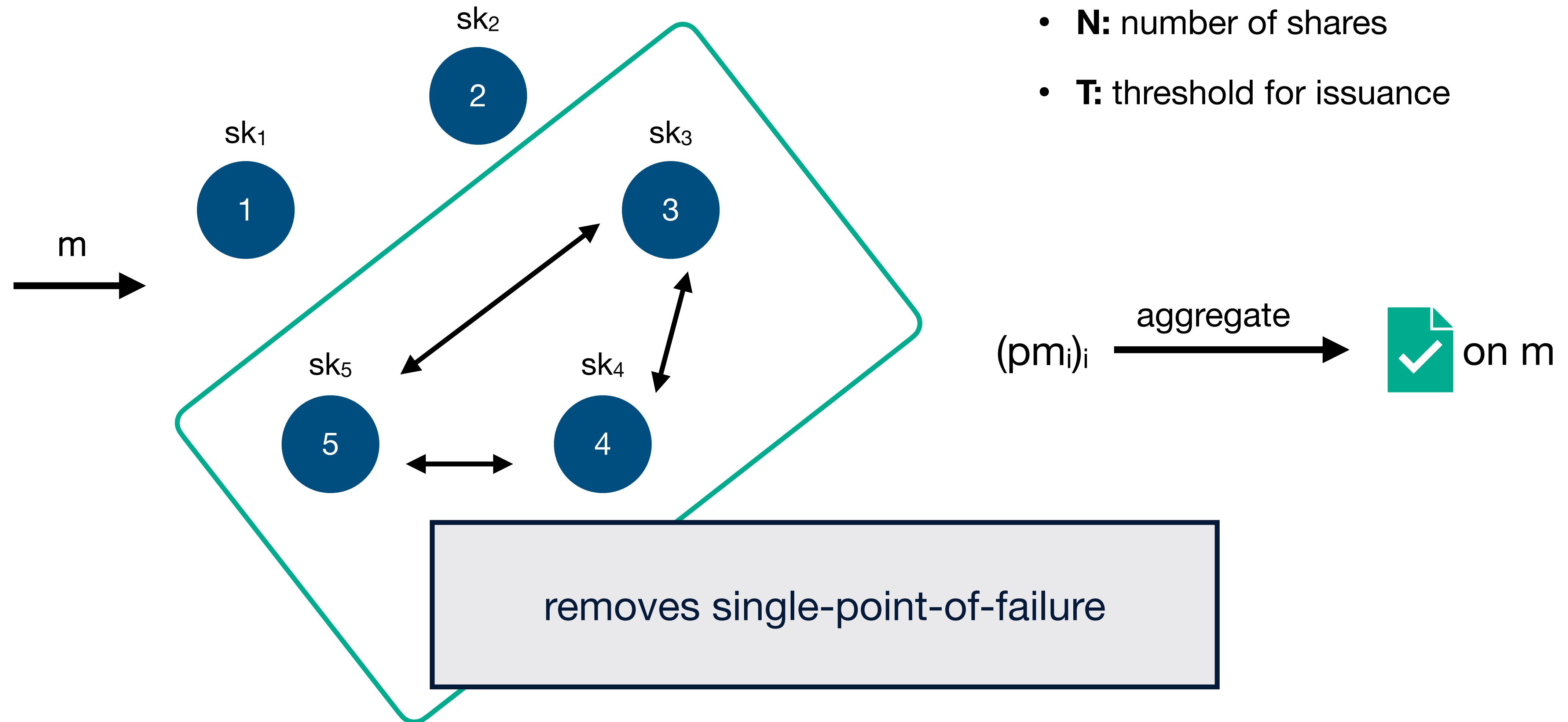
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Threshold Signatures

T-out-of-N

- **N**: number of shares
- **T**: threshold for issuance



Threshold Blind Signatures (TBS)

Blindness (User Privacy)

+

Threshold (Signer Security)

Threshold Blind Signatures (TBS)

Classical Setting

Snowblind: A Threshold Blind Signature in Pairing-Free Groups

Elizabeth Crites¹ , Chelsea Komlo² , Mary Maller³, Stefano Tessaro⁴ , and Chenzhi Zhu⁴ 

Stronger Security for Threshold Blind Signatures

Anja Lehmann¹ , Phillip Nazarian² , and Cavit Özbay¹ 

Threshold Blind Signatures from CDH

Michael Reichle and Zoé Reinke

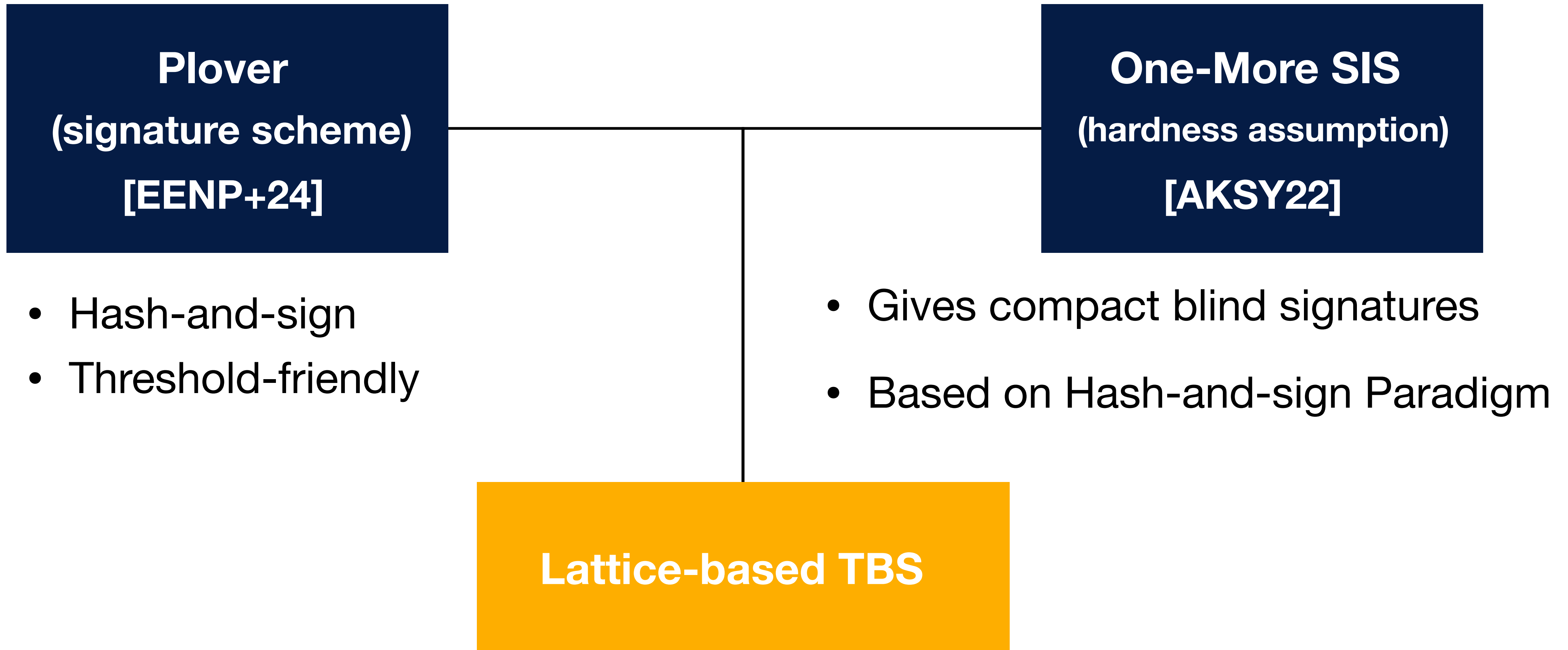
...

Post-Quantum Setting

**no candidate
(except for generic MPC)**

Question: How to achieve efficient TBS in the Post-Quantum setting?

Introducing the first Lattice-based TBS



Introducing the first Lattice-based TBS

Plover

1. Derive target = $H(\text{msg}, \text{salt})$
2. Find sig such that $\mathbf{A} \cdot \text{sig} = \text{target}$

[AKSY22]

1. **User** derives target = $H(\text{msg}) + \mathbf{A} \cdot \text{noise}$
2. **Server** finds sig such that $\mathbf{A} \cdot \text{sig} = \text{target}$
3. **User** computes final signature $\text{sig} - \text{noise}$.
Prove knowledge of final signature to preserve blindness

... many technical challenges to achieve provable security.

Introducing the first Lattice-based TBS

Plover

[AKSY22]

Targets uniform vs adversarially chosen

1. Derive $\text{target} = H(\text{msg}, \text{salt})$
2. Find sig such that $\mathbf{A} \cdot \text{sig} = \text{target}$

1. **User** derives $\text{target} = H(\text{msg}) + \mathbf{A} \cdot \text{noise}$
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[AKSY22]

Original reduction for Gaussian signatures

1. Derive target = $H(\text{msg}, \text{salt})$
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1. **User** derives target = $H(\text{msg}) + \mathbf{A} \cdot \text{noise}$
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[AKSY22]

*Original reduction for **statistically** uniform matrix A*

1. Derive target = $H(\text{msg}, \text{salt})$
2. Find sig such that $A \cdot \text{sig} = \text{target}$

1. **User** derives target = $H(\text{msg}) + A \cdot \text{noise}$
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Introducing the first Lattice-based TBS

Set	Comm./party	Signature size
Compact <i>LaBRADOR NIZK</i>	46 KB	66 KB
Easy to deploy <i>LNP NIZK</i>	35 KB	115 KB

Note: uses heuristics for the instantiation

“Lattice-based Threshold Blind Signatures”

By Sebastian Faller, Guilhem Niot, Michael Reichle

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